

This document explains how to reproduce the experimental results reported in the project.

Step-by-Step Instructions

1. Clone the repository:

```
git clone https://github.com/CrawfoBP23/ISE_fairness_project.git
```

2. Navigate to the project folder:

```
cd ISE_fairness_project
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Run the fairness testing experiment:

```
python fairness_tool.py
```

This will generate:

- fairness_results.csv
- fairness_summary.csv

5. Generate the graph:

```
python plot_graph.py
```

This will generate:

- idi_plot.png

Verification

- The fairness_summary.csv file should contain IDI ratios similar to those reported in the paper.
- The idi_plot.png graph should show a stable trend across budgets (100–1000).

Notes

- Results may vary slightly due to randomness, but overall trends should remain consistent.
- The experiment uses 30 repetitions per budget to reduce variance.